

KUKA KR 6 R900 sixx

KR AGILUS Family · 6-DOF Industrial Arm

ENGR 431 Final Project · Janga Bliss · Fort Lewis College

KR 6 R900 sixx

KR AGILUS Family

6 DOF

all revolute joints

0.9 m

reach

6 kg

payload

52 kg

arm weight

±0.03 mm

repeatability

Spherical wrist

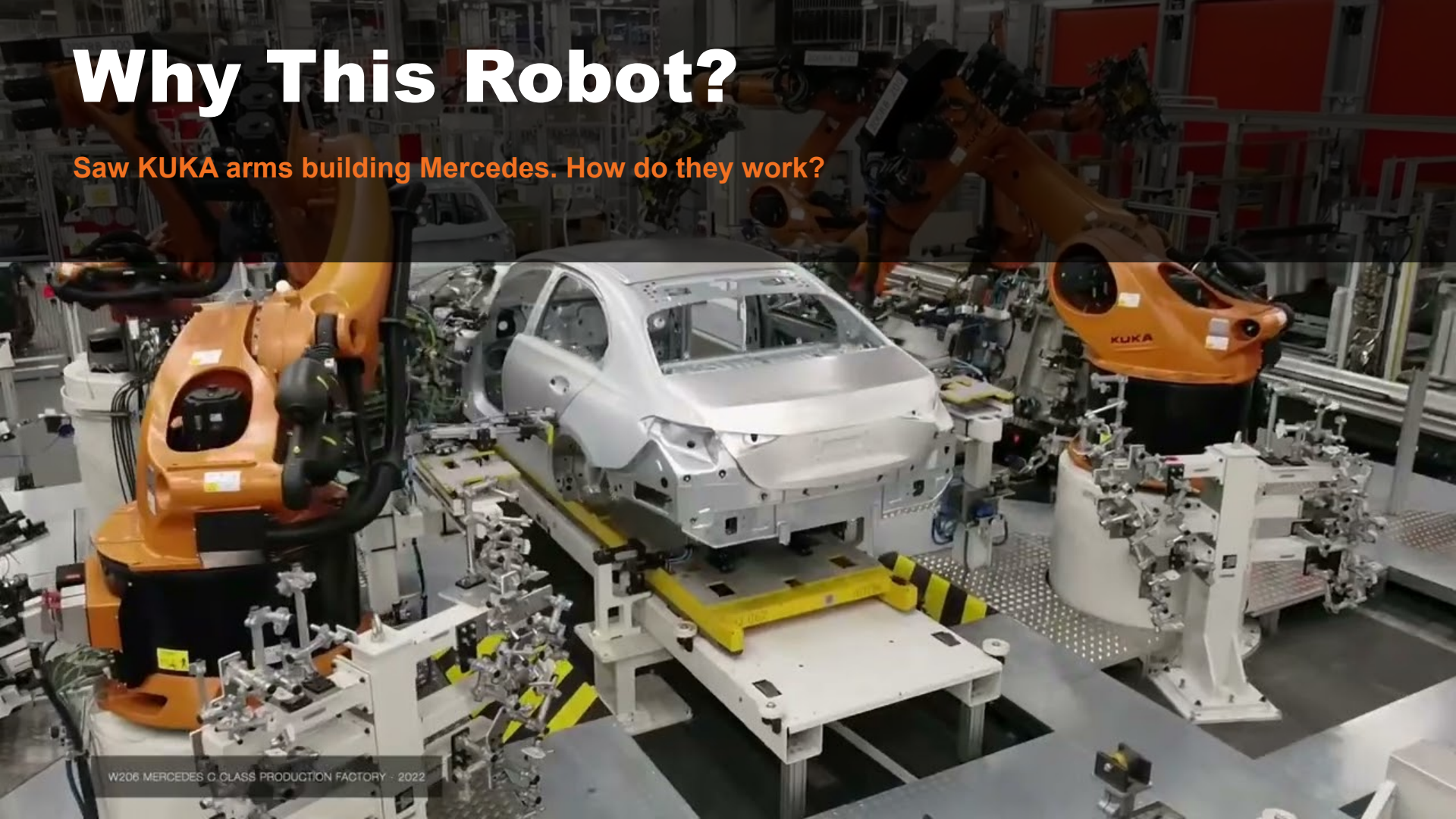
axes 4-5-6 intersect

Floor · Ceiling · Wall — Welding · Assembly · Handling

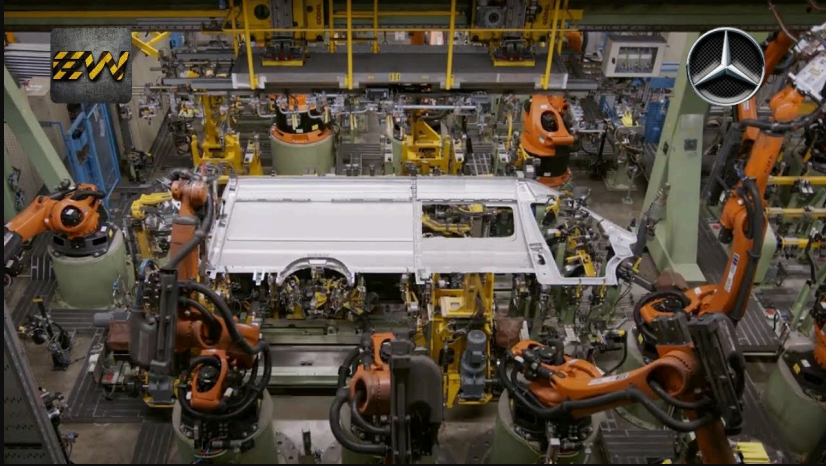


Why This Robot?

Saw KUKA arms building Mercedes. How do they work?



The KR AGILUS at Work



Panel Handling

8+ arms position a full van side panel in perfect coordination, with no humans



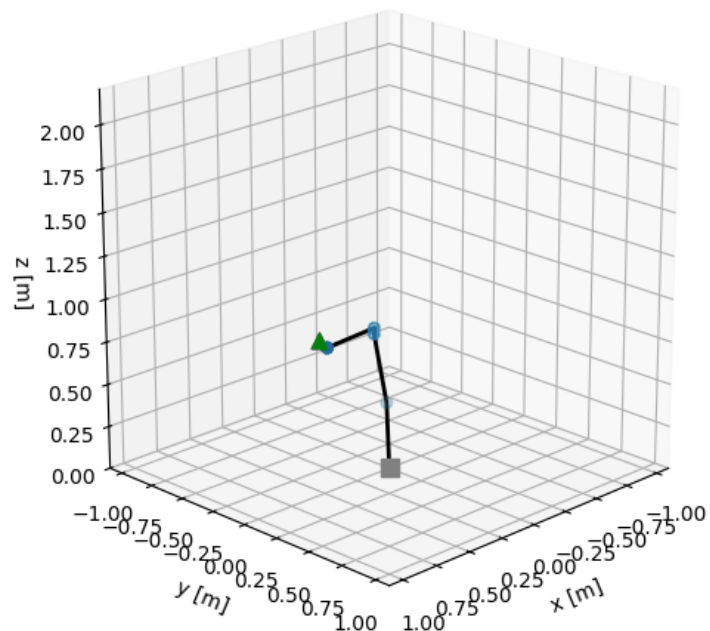
Welding Line

Tons of arms welding simultaneously sparks flying, sub-millimeter precision

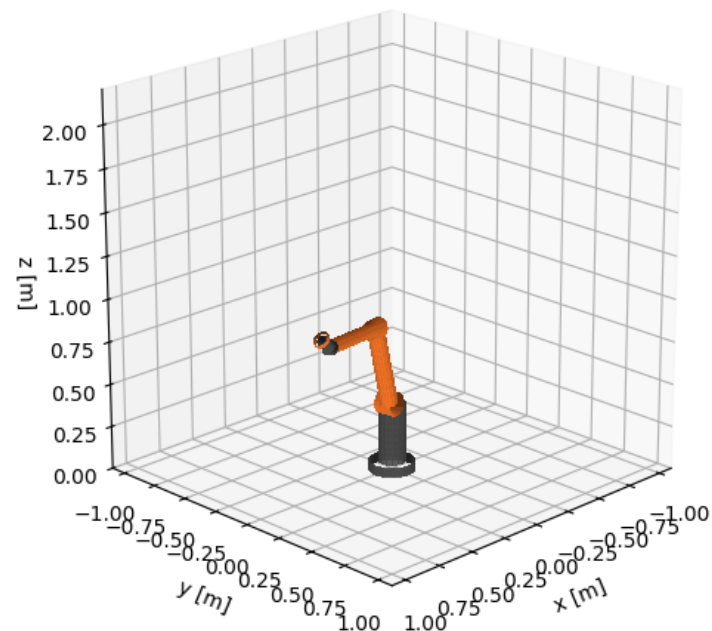
One factory can have **1,000+ KUKA robots** on a single production line with each one repeatable to ± 0.03 mm

My URDF Model

stick figure 1/200



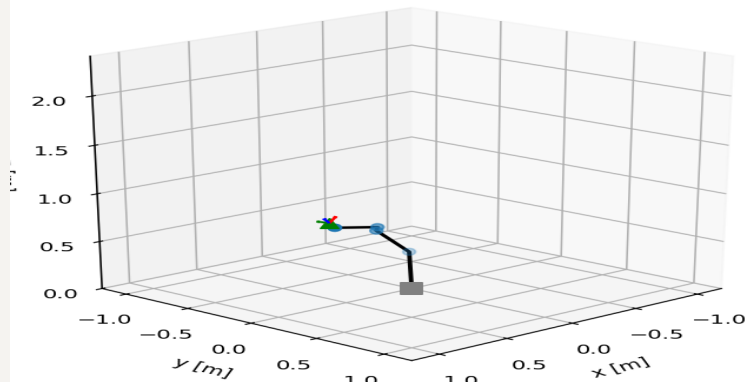
URDF



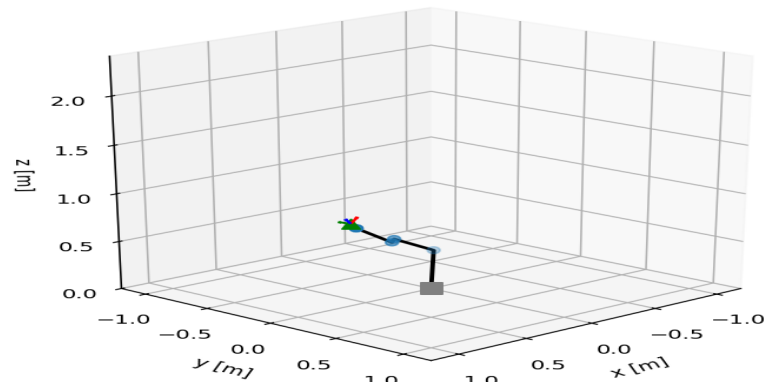
4 IK Branches

Same target · 4 different arm configurations

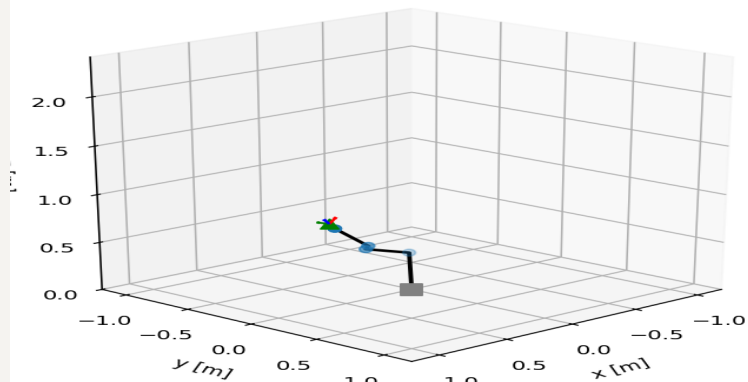
elbow up, shoulder front



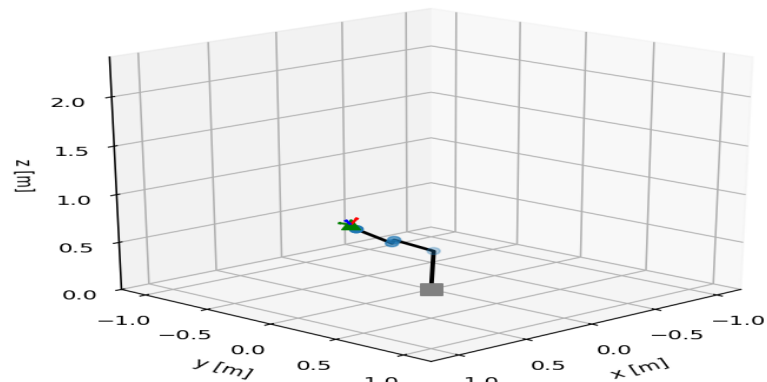
elbow up, shoulder back



elbow down, shoulder front



elbow down, shoulder back

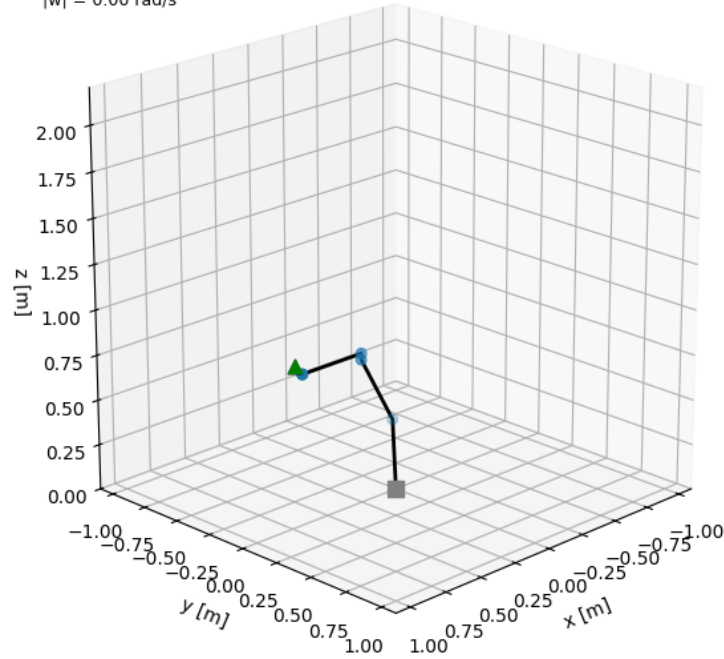


Velocity Arrows

v (magenta) · ω (cyan)

Part 5 (KR6): flange v (magenta) and w (cyan) (frame 1/80)

$|v| = 0.00 \text{ m/s}$
 $|w| = 0.00 \text{ rad/s}$



Thanks!

Jangara Bliss · ENGR 431 · Fort Lewis College